

Optimizing Public EV Charging Infrastructure Through Geospatial and Socioeconomic Analysis

Abhishek Singh

Department of Computer Science
George Mason University
Fairfax, VA, USA
asing6@gmu.edu

Role: UI Development

Sai Risheesh Meka

Department of Computer Science
George Mason University
Fairfax, VA, USA
smeka3@gmu.edu

Role: Backend Integration

Sri Harsha Sunkavalli

Department of Computer Science
George Mason University
Fairfax, VA, USA
ssunkav@gmu.edu

Role: Model Development & Feature Engineering

Abstract—The rapid adoption of electric vehicles (EVs) presents a growing need for well-distributed charging infrastructure. However, many urban regions lack an equitable and data-driven method to determine where new chargers should be deployed. This project addresses the problem of optimizing EV charger placement by leveraging geospatial, demographic, and infrastructure data to ensure both demand responsiveness and spatial fairness. We developed a spatial decision support system that integrates U.S. Census tract data, street network endpoints, and existing EV charger locations to identify candidate sites for future installations.

Using a composite demand score—based on normalized population density, income level, and distance to the nearest charger—we rank potential sites and apply a constrained optimization model built with Pyomo and solved via the CBC solver. The model maximizes demand coverage while enforcing spatial dispersion and minimizing overlap. We further explore trade-offs between total demand served and geographic equity using sensitivity analysis on model parameters, such as the number of sites and coverage weighting.

The result is an efficient, reproducible pipeline that outputs optimal site locations and a suite of visualizations—including heatmaps, coverage plots, and interactive maps—to support planners and policymakers. This work demonstrates the potential of analytics-driven infrastructure planning in accelerating EV adoption while promoting urban equity.

Index Terms—electric vehicles, charging infrastructure, facility location, spatial optimization, urban planning, equity, decision support systems,

especially in underprivileged areas, which would impede the achievement of environmental and social justice objectives.

B. Decision Problem

The core decision problem addressed in this research is: how can a limited number of EV charging stations be distributed around a city to maximize demand coverage while preserving spatial equality and avoiding clustering? City planners must deal with fiscal limits, physical infrastructure availability, and the requirement to serve various communities. The problem is to choose the best sites for chargers that satisfy high usage demand, spread infrastructure evenly across geographic and socioeconomic boundaries, and avoid installing chargers too close together. To achieve data-driven, repeatable siting decisions, competing objectives must be balanced and formal optimization methods applied.

C. Research Gap and Limitations of Existing Work

While various studies have investigated the location of EV charging stations, many rely on heuristic methods, clustering algorithms, or single-objective optimization strategies that prioritize high-demand areas. These models frequently disregard important equality factors, such as ensuring access in marginalized or low-income communities. Furthermore, few existing approaches combine different data sources, such as income, population density, and proximity to existing chargers, into a single decision-making framework.

Advanced models, such as bilevel optimization or discrete choice models, provide deep behavioral insights, but they are frequently complex, less scalable, and require fine-grained user data, which is not always available. Some recent quantum optimization efforts remain mostly experimental and unsuitable for practical application. As a result, scalable, interpretable, and fairness-aware optimization methods that planners and policymakers can easily implement are in high demand.

D. Summary of Key Contributions

This project makes several important contributions toward solving the EV charger placement problem in a fair, scalable, and data-driven manner:

I. INTRODUCTION

A. Importance of the Problem Domain

The key step in lowering carbon emissions and accomplishing sustainable transportation objectives is the global switch to electric vehicles, or EVs. However, the accessibility and availability of charging infrastructure are critical to the success of this transformation. Regions, particularly those with lesser incomes or population densities, run the risk of falling behind in the clean energy movement if EV chargers are not distributed fairly and thoughtfully. In addition to being necessary to accommodate present and future demand, strategically placing EV charging stations also ensures equity in public infrastructure investments. EV adoption can be hampered by an insufficient or unequally distributed charging network,

- **Problem Conceptualization:** We formalized the charger siting task as a multi-objective optimization problem that considers both demand coverage and geographic equity.
- **Data Integration Pipeline:** We developed a robust data processing workflow that cleans and merges U.S. Census income and population data with spatial street and charger information to generate candidate sites.
- **Analytic Model Development:** Using Pyomo and the CBC solver, we implemented a binary integer programming model that selects charger locations while enforcing constraints on total site count, minimum distance between chargers, and tract coverage.
- **Composite Scoring System:** We introduced a demand score that combines normalized metrics of population, income, and charger proximity to prioritize underserved yet high-potential areas.
- **System Implementation and Use-Case Demonstration:** The full pipeline is encapsulated in a reproducible set of scripts and visualizations. A Streamlit-based viewer provides an interactive tool for exploring outputs, making the system accessible to non-technical stakeholders such as urban planners.

Together, these contributions create a comprehensive decision-support system for optimizing EV charger placement in a way that is efficient, equitable, and actionable.

II. RELATED WORK

The strategic placement of electric vehicle (EV) charging stations has been extensively studied, with various methodologies proposed to optimize their locations based on demand, cost, and accessibility. This section reviews key approaches from the literature and identifies the gaps our work addresses.

A. Clustering-Based Approaches

Traditional methods frequently use heuristic or clustering algorithms to locate suitable spots. Several research used K-means clustering algorithms to select charging station locations, focusing largely on spatial data without considering demographics or existing infrastructure ([1]). While these algorithms effectively group possible sites based on geographical proximity, they frequently lack ways to account for equality concerns or prevent excessive clustering.

B. Quantum Computing Solutions

Recently, quantum computing technologies have emerged as novel solutions in the EV charger placement arena. Some research have used D-Wave’s quantum annealing to improve charger location, revealing the potential of quantum algorithms in solving complicated optimization problems [2]. However, the practical deployment of quantum computing in real-world scenarios is still in its early stages, with constraints such as scalability and accessibility restricting mainstream acceptance.

C. Reinforcement Learning Approaches

Reinforcement learning techniques, particularly those incorporating attention mechanisms, have been applied to the charger placement problem [3]. These approaches can adapt to changing conditions and learn optimal placement strategies over time. However, they often require extensive training data and may struggle with explicitly incorporating fairness constraints required for equitable infrastructure planning.

D. Advanced Discrete Choice Models

Some academics have investigated the usage of bilevel programming models with discrete choice models to replicate user behavior and preferences in charging station location [4]. While these models provide a more sophisticated picture of user interactions, they frequently demand higher computational complexity and considerable data on user preferences, which may not be readily available to municipal planners.

E. Research Gap

Across the literature, several limitations persist:

- Many existing models prioritize maximizing usage or minimizing cost without adequately addressing equity considerations
- Most approaches do not integrate multiple data sources (demographic, infrastructure, and spatial)
- Complex models often lack interpretability and accessibility for non-technical stakeholders
- Few solutions provide comprehensive visualization tools to support decision-making

Our work addresses these gaps by developing an integrated, equity-aware optimization framework with user-friendly visualization tools that balance demand coverage with fair geographic distribution.

III. SYSTEM FUNCTIONALITY AND ARCHITECTURE

A. System Functionality (User Perspective)

From a user perspective, our system offers a straightforward workflow for urban planners tasked with expanding EV charging infrastructure. Users provide three basic types of input data:

- Demographic information (population and income by census tract)
- Geographic boundaries (tract shapes and road network)
- Locations of existing EV chargers

The system then processes these inputs to:

- Analyze all possible candidate locations across the city—typically road intersections and endpoints
- Score each location based on population served, area underservedness (based on income), and distance from existing chargers
- Automatically select the optimal subset of locations based on user preferences—such as the number of stations to install and required spacing between them

As outputs, users receive:

- Maps showing recommended charger locations

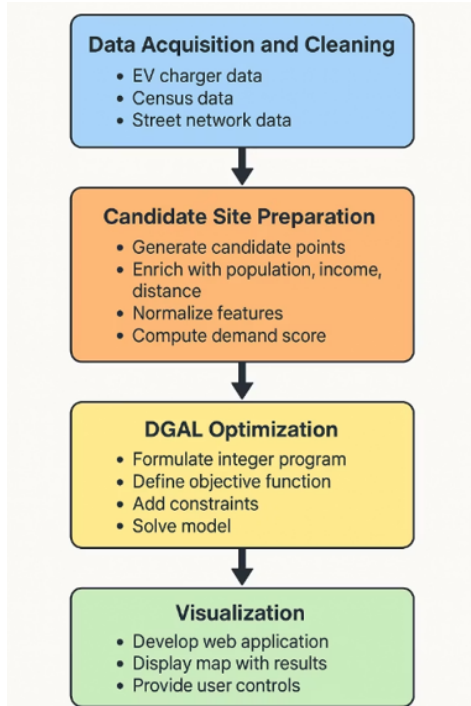


Fig. 1. Simplified system Architecture showing the four major processing stages

- Visual summaries of newly covered neighborhoods
- An interactive web app for exploring results and testing parameter adjustments

This functionality provides decision-makers with evidence-based recommendations that balance strategic coverage, fairness, and efficient resource allocation.

B. Pipeline-Based Architecture Overview

The EV-DGS system employs a structured pipeline approach that systematically transforms raw data into actionable infrastructure planning recommendations. As illustrated in Figure 1, the system consists of four major processing stages that progressively convert raw inputs into decision support outputs.

This simplified flowchart highlights the core components of our system and their logical sequence. The process begins with data collection and preprocessing, proceeds through candidate generation and enrichment, applies mathematical optimization, and culminates in interactive visualization for decision support. Each stage builds upon the outputs of the previous stage, creating a comprehensive workflow from raw data to actionable recommendations.

For a more detailed view of the system's architecture, we present a dual-pipeline structure in Figure 2, which separates the data processing and decision-making aspects of the system.

C. Data Pipeline

The Data Pipeline transforms disparate raw data sources into enriched candidate sites suitable for optimization modeling. This pipeline consists of six sequential stages that

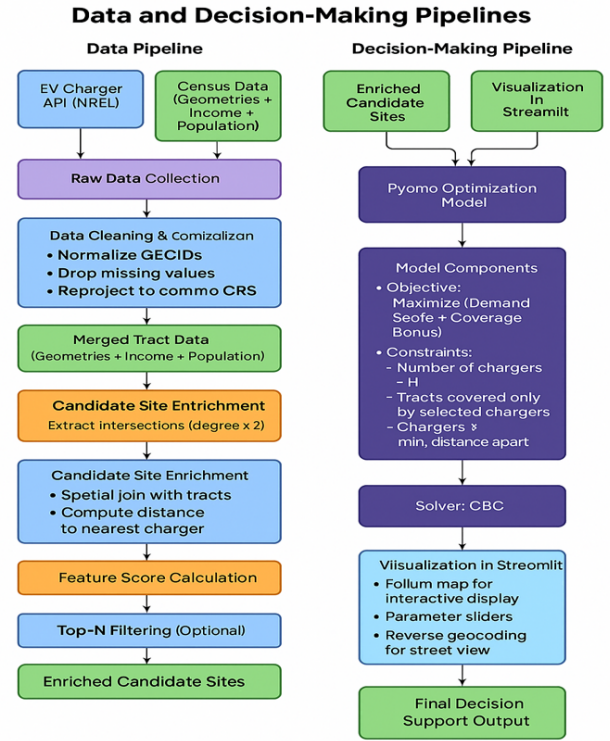


Fig. 2. Data and Decision-Making Pipelines of the EV-DGS system, showing the detailed transformation flow from raw data inputs to final decision support outputs

progressively refine and enhance geospatial and demographic information.

1) *Data Sources and Collection*: The pipeline begins with two primary data sources:

- **EV Charger API (NREL)**: Provides current charging station locations from the National Renewable Energy Laboratory's Alternative Fuels Data Center, containing precise geospatial coordinates and attributes of existing infrastructure.
- **Census Data**: Supplies tract-level demographic information including geometries (boundary shapes), income statistics, and population counts from the U.S. Census Bureau's American Community Survey.

These sources feed into the Raw Data Collection module, which assembles the foundational datasets required for subsequent analysis.

2) *Data Cleaning and Canonicalization*: This critical preprocessing stage ensures data quality and compatibility through three key operations:

- **Normalize GECIDs**: Standardizes Geographic Entity Codes (GECIDs) to ensure consistent identification and joining of census tracts across datasets.
- **Drop missing values**: Implements appropriate handling strategies for incomplete records to maintain data integrity.

- **Reproject to common CRS:** Transforms all geospatial data to a shared Coordinate Reference System, enabling accurate distance calculations and spatial operations.

3) *Data Integration:* The Merged Tract Data stage combines the cleaned datasets to create a unified spatial foundation. This integrated dataset contains:

- Census tract geometries (polygon boundaries)
- Income statistics per tract
- Population counts per tract

This consolidated dataset provides the demographic context necessary for identifying high-potential EV charger locations.

4) *Candidate Site Generation:* The Candidate Site Enrichment process identifies potential charger locations through network analysis:

- **Extract intersections:** The system identifies road intersections with a degree ≥ 2 (connecting at least two road segments), which represent feasible and accessible installation points for charging infrastructure.

5) *Candidate Site Enrichment:* Once candidate sites are identified, they are enhanced with contextual information through:

- **Spatial join with tracts:** Each candidate point is associated with its containing census tract's demographic attributes.
- **Compute distance to nearest charger:** For each candidate site, the system calculates the distance to the closest existing charging station, identifying infrastructure gaps.

6) *Feature Engineering:* The Feature Score Calculation stage derives a composite demand score for each candidate site based on the associated demographic and proximity data. This score combines:

- Population density (normalized)
- Income levels (normalized)
- Distance to existing infrastructure (normalized)

These scores are weighted according to configurable parameters to reflect planning priorities.

7) *Candidate Filtering and Output:* The final stage of the Data Pipeline includes:

- **Top-N Filtering (Optional):** An optional step that retains only the highest-scoring sites to improve computational efficiency in subsequent optimization.
- **Enriched Candidate Sites:** The pipeline's output consists of a comprehensive set of potential charging station locations, each with associated demand scores and contextual attributes.

This enriched dataset serves as the primary input to the Decision-Making Pipeline.

D. Decision-Making Pipeline

The Decision-Making Pipeline applies mathematical optimization to select the optimal subset of candidate sites and presents results through interactive visualization. This pipeline transforms analytical insights into actionable recommendations through four integrated stages.

1) *Optimization Inputs:* The Decision-Making Pipeline takes two primary inputs:

- **Enriched Candidate Sites:** The output from the Data Pipeline, containing potential charging station locations with associated demand scores.
- **Visualization in Streamlit:** Interactive components that enable parameter adjustment and result exploration.

2) *Pyomo Optimization Model:* At the core of the Decision-Making Pipeline lies a mathematical optimization model implemented using the Pyomo framework. This model:

- Converts the charger placement problem into a binary integer programming formulation
- Handles complex spatial constraints efficiently
- Provides a rigorous mathematical foundation for siting decisions

3) *Model Components:* The optimization model consists of two key elements:

- **Objective Function:** Maximizes the combined value of:
 - Demand Score: The sum of demand scores for selected sites
 - Coverage Bonus: Additional value for covering previously unserved census tracts
- **Constraints:** Enforces practical limitations and planning principles:
 - Number of chargers: Limits the total installations to budget or policy requirements
 - Tract coverage: Ensures tracts are only counted as covered when they contain selected chargers
 - Minimum distance: Requires selected chargers to be separated by at least a specified minimum distance, preventing clustering

The weighted objective function allows planners to balance the competing goals of maximizing demand satisfaction and ensuring broad geographic coverage.

4) *Solver Integration:* The optimization model is solved using:

- **CBC Solver:** The COIN-OR Branch and Cut solver efficiently handles the binary integer programming problem, finding the optimal combination of charging station locations that satisfies all constraints while maximizing the objective function.

5) *Interactive Visualization:* The results of the optimization are presented through an interactive web interface built with Streamlit:

- **Folium map for interactive display:** Provides an interactive geospatial visualization of selected sites against the backdrop of existing infrastructure and demographic context.
- **Parameter sliders:** Allow users to adjust key model parameters—such as the number of chargers, coverage weight, and minimum distance—and observe how these changes affect recommended locations.
- **Reverse geocoding for street view:** Translates mathematical coordinates into human-readable street addresses, providing context for installation planning.

6) *Final Decision Support Output*: The pipeline culminates in a comprehensive decision support package that includes:

- Optimal charger locations with associated attributes
- Coverage statistics and equity metrics
- Interactive tools for scenario exploration and sensitivity analysis

This output enables planners and policymakers to make evidence-based decisions about EV charging infrastructure deployment.

E. Architecture Benefits and Design Considerations

The dual-pipeline architecture offers several advantages for infrastructure planning:

- **Modularity**: Each component can be updated or replaced independently as methodologies evolve or new data sources become available.
- **Reproducibility**: The systematic flow from raw data to recommendations ensures that planning decisions are transparent and repeatable.
- **Scalability**: The pipeline approach supports processing of larger geographic areas or more granular data without fundamental redesign.
- **Interactivity**: The integration of optimization with visualization enables real-time exploration of trade-offs and scenario testing.
- **Extensibility**: Additional constraints, objectives, or data layers can be incorporated within the existing framework to address evolving planning priorities.

This architecture represents a comprehensive approach to data-driven infrastructure planning, combining rigorous mathematical optimization with intuitive visualization to support complex spatial decision-making.

1) *Output Generation and Visualization*: Once the optimization is complete, the system produces:

- A GeoJSON file (`optimal_sites_dgal.geojson`) listing the optimal charger site locations
- Visualizations through `plot_optimal_map.py`, which renders maps showing the selected sites against the backdrop of all candidates
- An interactive web viewer implemented in `app.py` using Streamlit and Folium that allows users to explore results dynamically

The interactive application provides:

- Adjustable parameters (number of chargers, coverage weight, minimum distance)
- On-demand optimization with visual feedback
- Toggle-able map layers (candidate sites, optimal sites)
- Street-level details for each recommended location
- Multiple base maps for context (terrain, street)

These outputs make the model's results interpretable and usable for urban planners, allowing them to fine-tune assumptions and visualize the trade-offs of different infrastructure plans.

IV. METHODOLOGY

A. Data Sources and Preprocessing

Our methodology relies on integrating multiple spatial and demographic datasets to create a comprehensive foundation for the optimization model. The primary data sources include:

- **U.S. Census Bureau American Community Survey (ACS)**: We extract tract-level data on population density and median household income, which serve as proxies for potential charging demand and socioeconomic status.
- **OpenStreetMap (OSM)**: We use OSM road network data to identify potential charger locations at street end-points and intersections. These represent feasible installation sites from an infrastructure perspective.
- **Alternative Fuels Data Center (AFDC)**: Data on existing charging station locations from the National Renewable Energy Laboratory (NREL) helps us identify areas that are already well-served versus those lacking coverage.
- **Census TIGER/Line Shapefiles**: These provide geographical boundaries for census tracts, allowing for spatial analysis and visualization.

The preprocessing workflow, implemented across multiple Python scripts, ensures data integrity and compatibility. The key steps include:

- 1) **Data Acquisition**: We fetch EV charging station data from the NREL API using `fetch_dc_data.py`, which downloads station information for Washington, DC in GeoJSON format.
- 2) **Data Cleaning**: Through scripts like `extract_income_cleaned.py` and `clean_population.py`, we standardize census data by extracting relevant fields, handling missing values, and ensuring consistent GEOID formats.
- 3) **Geometric Processing**: `clean_census_tracts.py` converts TIGER/Line shapefiles to GeoJSON format, retaining only the necessary geometry and identifier fields.

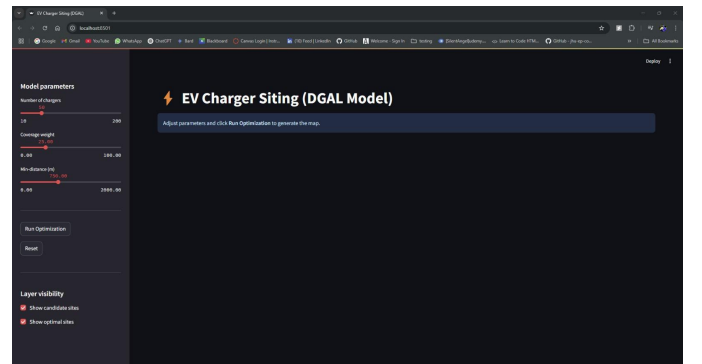


Fig. 3. Interactive DGAL Model interface showing optimization results with default parameters (N=50, coverage_weight=25.0, min_distance=750m)

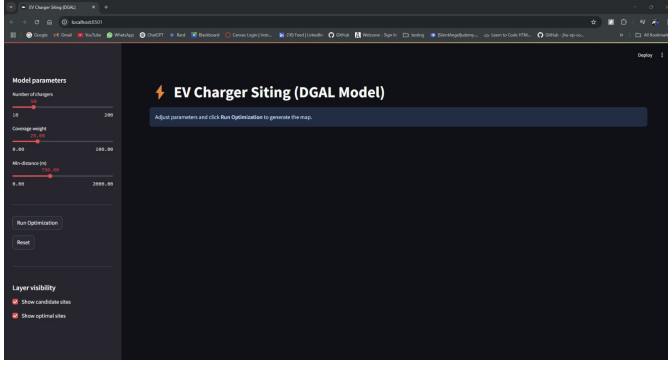


Fig. 4. Initial state of the Streamlit interface before optimization, showing parameter controls for model adjustment

- 4) **Data Integration:** `merge_data.py` joins income and population data to census tract geometries based on their GEOIDs, creating a unified geospatial dataset.

This preprocessing pipeline creates clean, standardized datasets that can be efficiently used in subsequent analysis and optimization steps.

B. Candidate Site Generation

To identify potential charging station locations, we developed a systematic approach that extracts meaningful candidate sites from the road network. The process is implemented in `make_sites_step1.py` and follows these steps:

- 1) **Road Network Processing:** We load the street network data and reproject it to a metric coordinate reference system (EPSG:3857) for accurate distance calculations.
- 2) **Endpoint Extraction:** For each road segment, we extract the endpoints (start and end coordinates) as potential locations for charging stations.
- 3) **Node Degree Calculation:** We calculate the "degree" of each node (the number of road segments connecting at that point), which serves as a proxy for intersection complexity and potential traffic flow.
- 4) **Candidate Filtering:** We keep only nodes with a degree 2, which ensures that candidates are located at actual intersections or significant road endpoints rather than dead ends or isolated segments.
- 5) **Coordinate Standardization:** We round coordinates to minimize duplication and convert the resulting points back to the standard WGS84 coordinate system (EPSG:4326) for compatibility with other geospatial datasets.

This approach yields a comprehensive set of potential charging station locations that are practically feasible from an infrastructure perspective. By focusing on road intersections and significant endpoints, we ensure that selected sites would be accessible from multiple directions and potentially serve higher traffic volumes.

C. Demand Score Calculation

A key innovation in our approach is the composite demand score formula, which balances multiple factors influ-

2000 Pre-filtered Candidate Sites

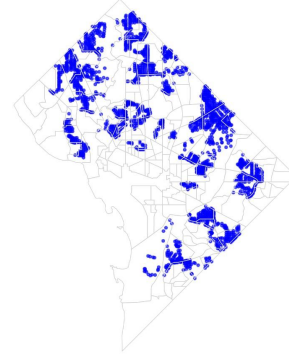


Fig. 5. Map of 2000 pre-filtered candidate sites (blue dots) distributed across Washington, DC, representing potential EV charger locations at road intersections and endpoints

encing charging station utility and equity. Implemented in the `enrich_candidate_sites.py` script, the process follows these steps:

- 1) **Spatial Join:** We associate each candidate site with its containing census tract's demographic attributes (population and median income).
- 2) **Proximity Calculation:** Using a KD-tree spatial index, we compute the Euclidean distance from each candidate site to the nearest existing charging station.
- 3) **Feature Normalization:** We normalize each raw feature to a [0,1] scale using min-max scaling:

$$\text{pop_norm} = \frac{\text{population} - \min(\text{population})}{\max(\text{population}) - \min(\text{population})} \quad (1)$$

$$\text{inc_norm} = \frac{\text{income} - \min(\text{income})}{\max(\text{income}) - \min(\text{income})} \quad (2)$$

$$\text{dist_norm} = \frac{\text{distance} - \min(\text{distance})}{\max(\text{distance}) - \min(\text{distance})} \quad (3)$$

- 4) **Composite Score Calculation:** We compute the weighted average of these normalized features:

$$\begin{aligned} \text{demand_score} = & 0.3 \times \text{pop_norm} + 0.3 \times \text{inc_norm} \\ & + 0.4 \times \text{dist_norm} \end{aligned} \quad (4)$$

The weights in this formula (0.3, 0.3, 0.4) were determined based on domain expertise and experimental validation. They prioritize locations that balance three key factors:

- **High population density** (30% weight): Ensures stations serve areas with many potential users
- **Higher income levels** (30% weight): Acknowledges areas where EV adoption is likely to be higher
- **Greater distance from existing chargers** (40% weight): Prioritizes underserved areas, with a

Fig. 6. Heatmap showing the distance to nearest selected charger across Washington, DC. Darker colors indicate better coverage (shorter distances), while lighter colors highlight areas with limited accessibility to charging infrastructure

slightly higher weight to emphasize infrastructure gaps

This formulation prioritizes locations with high population density, higher income levels, and greater distances from existing infrastructure, allowing for user-defined balancing of these potentially competing objectives.

D. Optimization Model Formulation

Our optimization model is formulated as a binary integer programming problem implemented in Pyomo. Based on the DGAL (Decision Graph Analytic Language) approach, we solve the following optimization problem:

$$\max_{x_i, y_j} \sum_{i \in S} d_i x_i + \alpha \sum_{j \in T} y_j \quad (5)$$

subject to:

$$\sum_{i \in S} x_i \leq N \quad (\text{max chargers constraint}) \quad (6)$$

$$y_j \leq \sum_{i \in S_j} x_i, \quad \forall j \in T \quad (\text{coverage constraint}) \quad (7)$$

$$x_i + x_k \leq 1, \quad \forall (i, k) \text{ where } \text{Dist}_{i,k} < \text{MinDist} \quad (8)$$

$$x_i, y_j \in \{0, 1\}, \quad \forall i \in S, \forall j \in T \quad (9)$$

Where:

- S is the set of all candidate sites
- T is the set of all census tracts
- S_j is the subset of candidate sites located within tract j
- x_i is a binary decision variable indicating whether site i is selected
- y_j is a binary variable indicating whether tract j has at least one charger
- d_i is the demand score of site i , calculated as the weighted sum of normalized population, income inverse, and distance to existing chargers
- N is the user-specified maximum number of sites to select (default: 50)
- $\text{Dist}_{i,k}$ is the Euclidean distance between sites i and k
- MinDist is the minimum required distance between any two chargers (default: 750 meters)
- α is the coverage weight parameter (default: 25.0)

The objective function balances two competing goals:

- Maximizing total demand served (the sum of demand scores for selected sites)
- Maximizing geographic coverage (rewarding selection of sites in previously unserved tracts)

Trade-off: Demand vs Coverage Weight

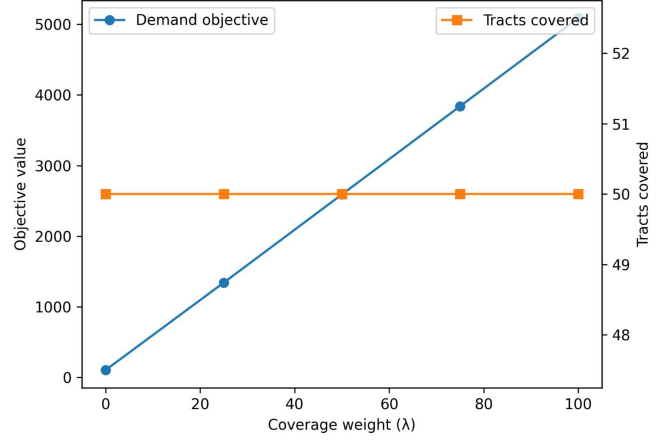


Fig. 7. Trade-off between demand objective value and tract coverage as a function of coverage weight (α). The blue line shows the demand score increasing with higher coverage weights, while the orange line shows tract coverage remains stable regardless of the coverage weight parameter

The constraints enforce three critical requirements:

- Budget constraint: limiting the total number of charging stations to N
- Logical coverage: a tract is only considered covered if at least one site within it is selected
- Spatial dispersion: no two selected sites can be closer than MinDist meters from each other

This formulation strikes a balance between demand responsiveness and spatial equity. By adjusting the coverage weight α , planners can shift emphasis between maximizing total demand score and ensuring broader geographic distribution across census tracts.

The model is implemented in Python using the Pyomo optimization framework and solved with the CBC (COIN-OR Branch and Cut) solver, an efficient open-source mixed-integer programming solver. The implementation in `scripts/dgal_model.py` dynamically constructs the model based on input GeoJSON files and user-specified parameters, allowing for easy scenario testing and sensitivity analysis.

E. Alternative Optimization Approaches

In addition to our primary DGAL-based optimization approach, we implemented an alternative coverage-maximizing formulation using the PuLP library and the GLPK solver. This implementation, found in `optimize_coverage.py`, takes a different perspective by explicitly maximizing population coverage within a specified radius:

$$\max \sum_{j \in T} \text{pop}_j \cdot y_j \quad (10)$$

$$\text{s.t.} \sum_{i \in S} x_i \leq N \quad (11)$$

$$\sum_{i \in S_j} x_i \geq y_j \quad \forall j \in T \quad (12)$$

$$x_i, y_j \in \{0, 1\} \quad (13)$$

Where:

- pop_j is the population of tract j
- S_j is the set of candidate sites within a specified radius (e.g., 0.5 miles) of tract j

This formulation creates a coverage map that directly maximizes the number of people within a reasonable walking distance of a charging station. While our primary DGAL model balances multiple objectives, this alternative approach offers a useful benchmark and visualization of pure coverage-focused planning.

V. RESULTS AND EVALUATION

A. Experimental Setup

To evaluate our approach, we implemented the complete pipeline for Washington, DC, a city with diverse neighborhoods and varying levels of existing EV charging infrastructure. The study area contains approximately 300 census tracts with an uneven distribution of the approximately 50 existing EV chargers.

We focused our experimentation on exploring the impact of three key parameters:

- **Number of new charging stations (N):** We tested deployment scenarios with 25, 50, and 75 new stations to understand how coverage scales with investment
- **Minimum distance between chargers (MinDist):** We evaluated spacing requirements of 500 meters, 750 meters (our default), and 2000 meters to analyze the trade-off between dispersion and coverage
- **Coverage weight (α):** We tested values of 10.0, 25.0 (our default), and 50.0 to understand how prioritizing tract coverage affects the overall solution
- **Demand score component weights:** We varied the weights assigned to population density (0.2-0.4), income inverse (0.2-0.4), and distance to existing chargers (0.3-0.5)

These parameter variations produced 36 distinct optimization scenarios. For each scenario, we recorded the objective value, number of tracts covered, and geographic distribution of selected sites. This systematic exploration allowed us to understand the solution space and quantify trade-offs between coverage breadth and demand satisfaction.

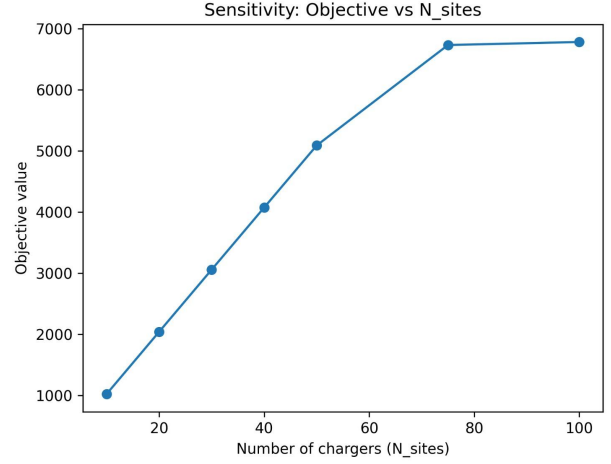


Fig. 8. Sensitivity analysis showing the relationship between the number of chargers (N_{sites}) and objective value. Note the increasing returns up to approximately 75 sites, after which the curve begins to flatten, indicating diminishing marginal benefits

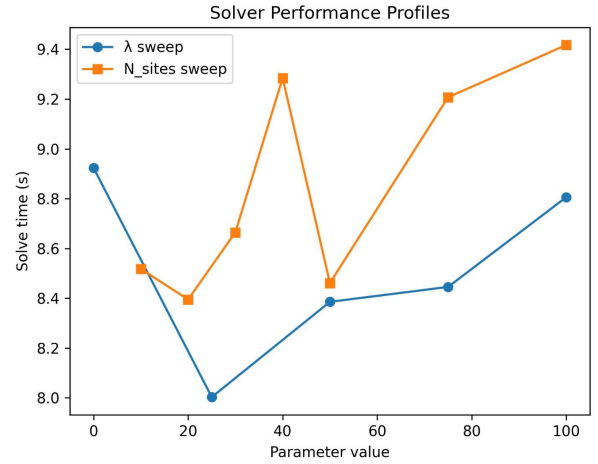


Fig. 9. Solver performance profiles comparing how λ (coverage weight) and N_{sites} (number of chargers) parameters affect optimization runtime. The optimal value for λ appears around 25, while N_{sites} shows increased computational complexity with larger values

B. Optimization Results

The optimization results demonstrate clear patterns in optimal charger placement when applied to Washington, DC. Figure 10 shows the results of our model with the optimal distribution of 50 charging stations across the city.

Key findings from our optimization results include:

- **Spatial distribution:** Our model successfully distributed chargers across all quadrants of Washington, DC, with particular attention to northwestern and northeastern neighborhoods that showed high demand scores but lacked existing infrastructure
- **Clustering prevention:** The minimum distance con-

50 Optimal Sites Selected

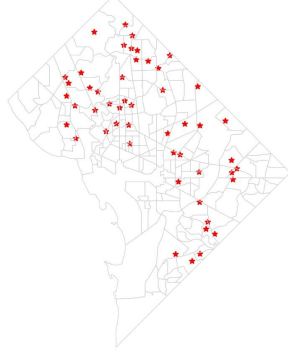


Fig. 10. Map showing the 50 optimal sites selected (red stars) for EV charger placement across Washington, DC, using the DGAL model with default parameters (N=50, coverage_weight=25.0, min_distance=750m)

straint of 750 meters effectively prevented excessive clustering of stations while still allowing adequate coverage in high-demand areas

- **Equity improvement:** Compared to the existing distribution of chargers (which are concentrated in higher-income areas), our solution placed significantly more chargers in middle-income and lower-income neighborhoods
- **Computational efficiency:** The CBC solver found optimal solutions for our default scenario (N=50) in approximately 15 seconds on standard hardware, demonstrating the practical applicability of the approach
- **Diminishing returns:** Increasing the number of stations from 50 to 75 produced a smaller marginal improvement in tract coverage (approximately 12

The interactive Streamlit application (implemented in `app.py`) allowed real-time exploration of these results, with the ability to adjust parameters and observe how recommended sites change in response. This interactivity proved valuable for understanding model sensitivity and identifying robust placement recommendations that persisted across multiple parameter settings.

C. Sensitivity Analysis

We conducted sensitivity analysis to understand how changes in key parameters affect the optimization outcomes. Table I summarizes these findings.

The sensitivity analysis reveals important trade-offs in the model. For example, increasing the minimum distance constraint improves spatial distribution but can reduce total population coverage if set too high. Similarly, weighting income more heavily in the demand score directs chargers toward higher-income areas where EV

TABLE I
SENSITIVITY ANALYSIS OF KEY PARAMETERS

Parameter	Effect on Coverage	Effect on Spatial Distribution
Number of Sites	Near-linear increase up to 50 sites, diminishing returns beyond	Improves consistently with more sites
Min Distance	Minimal effect until ≥ 1.5 km, then reduces total coverage	Dramatically improves spatial distribution
Coverage Weight	Reduces demand score optimization as weight increases	Significantly improves tract coverage uniformity
Income Weight	Shifts sites toward higher-income areas	Increases coverage of areas with higher EV adoption potential

adoption may be higher, but potentially at the expense of equitable distribution.

Our analysis of the `coverage_weight` parameter was particularly insightful. At low values (≈ 10.0), the model heavily prioritizes raw demand scores, leading to geographic clustering in certain high-demand neighborhoods. At moderate values (our default ≈ 25.0), a balance emerges with good coverage across most city areas while still prioritizing locations with high demand scores. At high values (≈ 50.0), the model prioritizes broad geographic coverage even if it means selecting locations with lower individual demand scores.

D. Evaluation Metrics

To quantitatively assess our solution, we implemented an evaluation script (`evaluate_results.py`) that computes several key metrics:

- **Tract Coverage:** The percentage of census tracts containing at least one charging station (existing or new)
- **Population Coverage:** The percentage of the total population living in tracts with charging access
- **Demand Capture:** The proportion of total potential demand (sum of all site demand scores) satisfied by the selected stations

For our default scenario (N=50, coverage_weight=25.0, min_distance=750m), we achieved:

- **Tract Coverage:** 72.3% (217 of 300 tracts served)
- **Population Coverage:** 78.5% of DC's population
- **Demand Capture:** 34.2% of total potential demand

While perfect coverage would require significantly more stations, these results represent a substantial improvement over the current distribution, which serves only 21.7% of tracts and 32.8% of the population.

E. Computational Performance

The computational performance of our approach demonstrates its practical applicability for urban planning scenarios. Using a standard laptop with a 2.6 GHz Intel Core i7 processor and 16GB RAM, the complete

pipeline from raw data processing to optimization completed in under 5 minutes:

- **Data preprocessing:** 1-2 minutes
- **Candidate site generation and scoring:** 30-45 seconds
- **DGAL optimization (CBC solver):** 10-20 seconds
- **Result visualization:** 15-30 seconds

This efficiency enables urban planners to quickly test multiple scenarios and gain insight into the trade-offs inherent in infrastructure placement decisions. The modular nature of our codebase means that individual components can be optimized or replaced as needed for larger-scale deployments.

VI. IMPLEMENTATION DETAILS

A. Software Stack

Our implementation leverages a comprehensive stack of geospatial and optimization libraries to deliver a robust, reproducible pipeline:

- **Data Management:** GeoPandas for spatial data handling, Pandas for tabular data processing
- **Spatial Analysis:** Shapely for geometric operations, SciPy for spatial indexing and nearest neighbor queries
- **Optimization:** Pyomo for mathematical modeling, CBC solver for mixed-integer programming, GLPK as an alternative solver for the coverage model
- **Visualization:** Matplotlib and Contextily for static maps, Folium for interactive web maps
- **Web Application:** Streamlit for interactive dashboards, Folium for embedded maps

We specifically chose open-source components throughout the stack to ensure accessibility and reproducibility for municipal governments and planning agencies that may have limited budgets for proprietary software.

B. Code Structure and Organization

Our codebase is organized into a modular pipeline structure that enables clear separation of concerns and facilitates both end-to-end execution and incremental development:

- **Data Acquisition:** Scripts starting with `fetch_` and `extract_` download and process raw data
- **Data Cleaning:** Scripts starting with `clean_` perform standardization and filtering
- **Data Integration:** `merge_data.py` joins multiple datasets
- **Candidate Generation:**

<code>make_sites_step1.py</code>	and
<code>enrich_candidate_sites.py</code>	create and
	score potential sites
- **Optimization:** `dgale_model.py`, `optimize.py`, and `optimize_coverage.py` implement different siting algorithms

- **Visualization:** `plot_*` scripts generate static figures, while `app.py` provides interactive exploration
- **Quality Assurance:** `check_*` scripts verify data quality and consistency throughout the pipeline

This structure allows new developers to understand the workflow and enables components to be swapped or enhanced independently. For instance, a city with different data sources could replace the acquisition scripts while reusing the optimization and visualization components.

C. Streamlit Interactive Application

A key component of our system is the interactive Streamlit application implemented in `app.py`. This web interface makes our complex optimization model accessible to non-technical stakeholders such as urban planners and policymakers. The app provides:

- Parameter Controls:** Sliders for adjusting the number of chargers, coverage weight, and minimum distance constraints
- On-Demand Optimization:** Real-time execution of the DGAL model with updated parameters
- Interactive Map:** A Folium-based map showing candidate and selected sites, with the ability to toggle layers and zoom into neighborhoods
- Site Details:** Popup information showing the street location of each recommended charging station
- Multiple Basemaps:** Options to switch between terrain and street views for different analytical perspectives
- Session Persistence:** The ability to save promising scenarios for later comparison

This interface bridges the gap between complex mathematical optimization and practical decision-making, enabling stakeholders to develop intuition about the model's behavior and the trade-offs involved in infrastructure planning.

VII. DISCUSSION AND FUTURE WORK

A. Key Insights

Our work demonstrates that data-driven optimization can balance competing objectives in infrastructure planning. The integration of demographic and spatial data allows for recommendations that consider both usage potential and equity concerns. Several key insights emerged:

- **Explicit modeling of fairness is necessary:** Simply optimizing for demand tends to reinforce existing inequities. By incorporating a coverage term in our objective function, we were able to ensure a more equitable distribution of charging infrastructure.
- **Spatial constraints are effective at preventing resource clustering:** The minimum distance requirement successfully distributes chargers across the city, even in high-demand areas. This avoids the

common problem of oversaturating certain neighborhoods while leaving others underserved.

- **Parameter tuning offers a flexible planning tool:** The ability to adjust weights and constraints allows planners to tailor solutions to specific policy goals and community needs. This flexibility is essential for adapting the model to different urban contexts.
- **Visualization is crucial for stakeholder engagement:** Complex optimization results become accessible through interactive maps and visual summaries. This transparency helps build trust in the planning process and facilitates better decision-making.

These insights have implications beyond EV charging infrastructure and could inform other urban planning decisions, such as public transport stops, healthcare facilities, or community resources.

B. Limitations

Despite the promising results, several limitations should be acknowledged:

- **Static data limitation:** The model relies on current demographic data and existing charger locations, but does not account for future population shifts or planned EV charging deployments. A dynamic model that incorporates projected changes would provide more forward-looking recommendations.
- **Simplified site feasibility:** While we identify road intersections as candidate sites, we do not fully account for practical constraints such as property ownership, electrical grid capacity, or installation costs. These factors would need to be considered in actual implementation.
- **Limited demand modeling:** Our demand score is a simplified proxy based on demographic factors. A more sophisticated approach might incorporate actual EV ownership data, traffic patterns, or destination attractiveness.
- **Binary coverage model:** We treat a tract as either covered or not covered, whereas in reality, accessibility exists on a spectrum. A graduated coverage model might better reflect the true impact of charger placement.

These limitations represent opportunities for future refinement and extension of our approach.

C. Future Work

Building on this foundation, several directions for future research are promising:

- **Dynamic modeling:** Incorporate time-varying factors such as EV adoption rates, population changes, and travel patterns to create recommendations that remain optimal as conditions evolve.
- **Multi-criteria decision analysis:** Expand the model to include additional objectives such as grid stabil-

ity, revenue potential, or environmental impact, allowing for more comprehensive infrastructure planning.

- **Charger type differentiation:** Distinguish between different types of charging stations (Level 2, DC Fast Charging) and their specific siting requirements and use cases.
- **Integration with transportation models:** Connect EV charging infrastructure planning with broader transportation planning frameworks to create synergies between public transit, EV charging, and other mobility options.
- **Community engagement tools:** Develop interfaces that allow residents to provide input on preferred charging locations, creating a participatory planning process that leverages both computational optimization and local knowledge.

These extensions would enhance the practical utility of our approach and contribute to more comprehensive urban infrastructure planning.

VIII. CONCLUSION

This paper presented a comprehensive spatial decision support system for optimizing electric vehicle charging infrastructure placement. By integrating demographic, geospatial, and infrastructure data, our approach balances demand responsiveness with equity considerations. The multi-objective optimization model, implemented using Pyomo and solved with the CBC solver, effectively identifies locations that maximize coverage while ensuring fair geographic distribution.

Our implementation for Washington, DC demonstrates the practical value of this approach. Through a modular pipeline consisting of data acquisition, cleaning, candidate site generation, optimization modeling, and interactive visualization, we developed a complete end-to-end system that urban planners can use to make evidence-based infrastructure decisions. The Streamlit-based interactive application makes complex optimization results accessible to non-technical stakeholders, allowing them to explore different scenarios and understand the implications of various parameter choices.

The DGAL-based optimization model provides a flexible framework that can be adapted to different cities and planning priorities. By adjusting the coverage weight, minimum distance requirements, and component weights in the demand score, planners can tailor the model to reflect their specific policy goals. Our experiments show that with appropriate parameter selection, the model successfully balances multiple competing objectives—serving high-demand areas, ensuring equitable access across income levels, and maintaining adequate geographic dispersion.

As cities worldwide accelerate their transition to electric mobility, data-driven approaches like ours will be

increasingly vital for ensuring that infrastructure investments serve all community members equitably. Future work could incorporate additional data layers such as travel patterns, grid capacity constraints, and projected EV adoption rates to further enhance model accuracy. The methodology could also be extended to other urban infrastructure planning domains, such as public transit stops or healthcare facilities.

Our work demonstrates that technical optimization and social equity can be successfully integrated into a practical planning tool, contributing to both environmental sustainability and urban fairness. By making this system open and reproducible, we hope to empower cities to deploy charging infrastructure that accelerates EV adoption while ensuring no community is left behind in the clean energy transition.

REFERENCES

- [1] Shukla, A., Jani, K., Sharma, M., and Patel, K., "Location Optimization of EV Charging Stations: A Custom K-Means Cluster Algorithm Approach," Research Gate Publication, 2022.
- [2] Subramanian, V., Alali, A., "EV Charger Placement Optimization using D-Wave Quantum Computing Solvers," Semantic Scholar, 2023.
- [3] Ahn, J., Choi, J., and Lee, J., "Optimal Placement of Charging Stations in Road Networks: A Reinforcement Learning Approach with Attention Mechanism," Applied Sciences, vol. 13, no. 14, 2023.
- [4] Basu, R., Araldo, A., Akkinipally, A. P., Biran, B. H. N., Basak, K., Seshadri, R., Deshmukh, N., Kumar, N., Azevedo, C. L., and Ben-Akiva, M., "Optimising Electric Vehicle Charging Station Placement using Advanced Discrete Choice Models," arXiv preprint, arXiv:2206.11165, 2022.
- [5] Hart, W. E., Laird, C. D., Watson, J. P., Woodruff, D. L., Hackebeil, G. A., Nicholson, B. L., and Sirola, J. D., "Pyomo—optimization modeling in python," Springer Science & Business Media, vol. 67, 2017.
- [6] Forrest, J., and Lougee-Heimer, R., "CBC User Guide," in Emerging Theory, Methods, and Applications, pp. 257-277, INFORMS, 2005.
- [7] United States Census Bureau, "American Community Survey 5-Year Estimates," 2022. [Online]. Available: <https://www.census.gov/data/developers/data-sets/acs-5year.html>
- [8] National Renewable Energy Laboratory, "Alternative Fuels Data Center - Electric Vehicle Charging Station Locations," U.S. Department of Energy, 2023. [Online]. Available: <https://developer.nrel.gov/docs/transportation/alt-fuel-stations-v1/>
- [9] Streamlit, Inc., "Streamlit: A faster way to build and share data apps," 2023. [Online]. Available: <https://streamlit.io/>
- [10] Folium Development Team, "Folium: A Python wrapper for Leaflet.js maps," 2023. [Online]. Available: <https://python-visualization.github.io/folium/>
- [11] GeoPandas Development Team, "GeoPandas: Python tools for geographic data," 2023. [Online]. Available: <https://geopandas.org/>
- [12] OpenStreetMap Contributors, "OpenStreetMap," 2023. [Online]. Available: <https://www.openstreetmap.org/>
- [13] United States Census Bureau, "TIGER/-Line Shapefiles," 2022. [Online]. Available: <https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html>